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EXPERIMENT 11

Exp No:11 Experiment to understand pandas library use cases in Data science.

AIM:
To understand pandas library use cases in Data science.

DESCRIPTION:
Understand data frame use cases using pandas library

CODE:

```
import numpy as np

import pandas as pd

list = [

    [101, 'Maha', 45000],

    [102, 'Sweta', 52000],

    [103, 'Keerthana', 68000],

    [104, 'Rakshana', 52000],

    [105, 'Nekha', 75000]

]

df = pd.DataFrame(list)

print(df)

df.columns = ['Empd', 'Name', 'Salary']

print(df)

df.info()

# Read your actual dataset

df = pd.read_csv("/content/Employee_Salary_Dataset.csv")
```

```
print(df)
df.info()
print(df.head())
print(df.tail())
# Salary column in your dataset is "Salary"
print(df.Salary)
print(type(df.Salary))
print(df.Salary.mean())
print(df.Salary.median())
print(df.Salary.mode())
print(df.Salary.var())
print(df.Salary.std())
print(df.describe())
Print(df.describe(include='all'))
empCol = df.columns
print(empCol)
emparray = df.values
print(emparray)
employee_DF = pd.DataFrame(emparray, columns=empCol)
print(employee_DF)
```

OUTPUT:

```
0    1    2
0  101    Maha  45000
1  102    Sweta  52000
```

2 103 Keerthana 68000

3 104 Rakshana 52000

4 105 Nekha 75000

Empd Name Salary

0 101 Maha 45000

1 102 Sweta 52000

2 103 Keerthana 68000

3 104 Rakshana 52000

4 105 Nekha 75000

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 5 entries, 0 to 4

Data columns (total 3 columns):

Column Non-Null Count Dtype

--- ------ -------------- -----

0 Empd 5 non-null int64

1 Name 5 non-null object

2 Salary 5 non-null int64

dtypes: int64(2), object(1)

memory usage: 252.0+ bytes

ID Experience_Years Age Gender Salary

0 1 5 28 Female 250000

1 2 1 21 Male 50000

2 3 3 23 Female 170000

3 4 2 22 Male 25000

4 5 1 17 Male 10000

5 6 25 62 Male 5001000

6 7 19 54 Female 800000

7 8 2 21 Female 9000

8	9	10	36	Female	61500
9	10	15	54	Female	650000
10	11	4	26	Female	250000
11	12	6	29	Male	1400000
12	13	14	39	Male	6000050
13	14	11	40	Male	220100
14	15	2	23	Male	7500
15	16	4	27	Female	87000
16	17	10	34	Female	930000
17	18	15	54	Female	7900000
18	19	2	21	Male	15000
19	20	10	36	Male	330000
20	21	15	54	Male	6570000
21	22	4	26	Male	25000
22	23	5	29	Male	6845000
23	24	1	21	Female	6000
24	25	4	23	Female	8900
25	26	3	22	Female	20000
26	27	1	18	Male	3000
27	28	27	62	Female	10000000
28	29	19	54	Female	5000000
29	30	2	21	Female	6100
30	31	10	34	Male	80000
31	32	15	54	Male	900000
32	33	20	55	Female	1540000
33	34	19	53	Female	9300000
34	35	16	49	Male	7600000

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 35 entries, 0 to 34

Data columns (total 5 columns):

#	Column	Non-Null Count	Dtype
0	ID	35 non-null	int64
1	Experience_Years	35 non-null	int64
2	Age	35 non-null	int64
3	Gender	35 non-null	object
4	Salary	35 non-null	int64

dtypes: int64(4), object(1)

memory usage: 1.5+ KB

	ID	Experience_Years	Age	Gender	Salary
0	1	5	28	Female	250000
1	2	1	21	Male	50000
2	3	3	23	Female	170000
3	4	2	22	Male	25000
4	5	1	17	Male	10000

	ID	Experience_Years	Age	Gender	Salary
30	31	10	34	Male	80000
31	32	15	54	Male	900000
32	33	20	55	Female	1540000
33	34	19	53	Female	9300000
34	35	16	49	Male	7600000

0 250000

1 50000

2 170000

3 25000

4 10000

5	5001000
6	800000
7	9000
8	61500
9	650000
10	250000
11	1400000
12	6000050
13	220100
14	7500
15	87000
16	930000
17	7900000
18	15000
19	330000
20	6570000
21	25000
22	6845000
23	6000
24	8900
25	20000
26	3000
27	10000000
28	5000000
29	6100
30	80000
31	900000
32	1540000

```
33     9300000
34     7600000
Name: Salary, dtype: int64
<class 'pandas.core.series.Series'>
2059147.142857143
250000.0
0     25000
1    250000
Name: Salary, dtype: int64
10049683866903.361
3170123.635901818
           ID  Experience_Years        Age        Salary
count  35.000000          35.00000  35.000000  3.500000e+01
mean   18.000000           9.20000  35.485714  2.059147e+06
std    10.246951           7.55295  14.643552  3.170124e+06
min     1.000000           1.00000  17.000000  3.000000e+03
25%     9.500000           2.50000  22.500000  2.250000e+04
50%    18.000000           6.00000  29.000000  2.500000e+05
75%    26.500000          15.00000  53.500000  3.270000e+06
max    35.000000          27.00000  62.000000  1.000000e+07
           ID  Experience_Years        Age  Gender        Salary
count   35.000000          35.00000  35.000000      35  3.500000e+01
unique        NaN              NaN       NaN       2           NaN
top           NaN              NaN       NaN  Female           NaN
freq          NaN              NaN       NaN      18           NaN
mean    18.000000           9.20000  35.485714     NaN  2.059147e+06
std     10.246951           7.55295  14.643552     NaN  3.170124e+06
min      1.000000           1.00000  17.000000     NaN  3.000000e+03
```

25% 9.500000 2.50000 22.500000 NaN 2.250000e+04

50% 18.000000 6.00000 29.000000 NaN 2.500000e+05

75% 26.500000 15.00000 53.500000 NaN 3.270000e+06

max 35.000000 27.00000 62.000000 NaN 1.000000e+07

Index(['ID', 'Experience_Years', 'Age', 'Gender', 'Salary'], dtype='object')

[[1 5 28 'Female' 250000]

[2 1 21 'Male' 50000]

[3 3 23 'Female' 170000]

[4 2 22 'Male' 25000]

[5 1 17 'Male' 10000]

[6 25 62 'Male' 5001000]

[7 19 54 'Female' 800000]

[8 2 21 'Female' 9000]

[9 10 36 'Female' 61500]

[10 15 54 'Female' 650000]

[11 4 26 'Female' 250000]

[12 6 29 'Male' 1400000]

[13 14 39 'Male' 6000050]

[14 11 40 'Male' 220100]

[15 2 23 'Male' 7500]

[16 4 27 'Female' 87000]

[17 10 34 'Female' 930000]

[18 15 54 'Female' 7900000]

[19 2 21 'Male' 15000]

[20 10 36 'Male' 330000]

[21 15 54 'Male' 6570000]

[22 4 26 'Male' 25000]

[23 5 29 'Male' 6845000]

[24 1 21 'Female' 6000]

[25 4 23 'Female' 8900]

[26 3 22 'Female' 20000]

[27 1 18 'Male' 3000]

[28 27 62 'Female' 10000000]

[29 19 54 'Female' 5000000]

[30 2 21 'Female' 6100]

[31 10 34 'Male' 80000]

[32 15 54 'Male' 900000]

[33 20 55 'Female' 1540000]

[34 19 53 'Female' 9300000]

[35 16 49 'Male' 7600000]]

	ID	Experience_Years	Age	Gender	Salary
0	1	5	28	Female	250000
1	2	1	21	Male	50000
2	3	3	23	Female	170000
3	4	2	22	Male	25000
4	5	1	17	Male	10000
5	6	25	62	Male	5001000
6	7	19	54	Female	800000
7	8	2	21	Female	9000
8	9	10	36	Female	61500
9	10	15	54	Female	650000
10	11	4	26	Female	250000
11	12	6	29	Male	1400000
12	13	14	39	Male	6000050
13	14	11	40	Male	220100
14	15	2	23	Male	7500

15	16	4	27	Female	87000
16	17	10	34	Female	930000
17	18	15	54	Female	7900000
18	19	2	21	Male	15000
19	20	10	36	Male	330000
20	21	15	54	Male	6570000
21	22	4	26	Male	25000
22	23	5	29	Male	6845000
23	24	1	21	Female	6000
24	25	4	23	Female	8900
25	26	3	22	Female	20000
26	27	1	18	Male	3000
27	28	27	62	Female	10000000
28	29	19	54	Female	5000000
29	30	2	21	Female	6100
30	31	10	34	Male	80000
31	32	15	54	Male	900000
32	33	20	55	Female	1540000
33	34	19	53	Female	9300000
34	35	16	49	Male	7600000

RESULT:

Thus, the Pandas library was used successfully to create, manipulate, and analyze DataFrames. Various operations such as data inspection, statistical calculations, and accessing rows and columns were performed successfully.